

Assignment 2

Week 2

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Problem 1. Normalization by Hand

Normalize vector $|\vec{v}_1\rangle = 2|\uparrow\rangle + 1|\downarrow\rangle$ by hand.

Solution. To normalize the vector $|\vec{v}_1\rangle = 2|\uparrow\rangle + 1|\downarrow\rangle$, we need to divide it by its norm. The norm of $|\vec{v}_1\rangle$ is given by:

$$||\vec{v}_1\rangle| = \sqrt{|2|^2 + |1|^2} = \sqrt{4 + 1} = \sqrt{5}$$

Therefore, the normalized vector is:

$$|\vec{v}_{1\text{normalized}}\rangle = \frac{2}{\sqrt{5}}|\uparrow\rangle + \frac{1}{\sqrt{5}}|\downarrow\rangle \approx \begin{pmatrix} 0.8944 \\ 0.4472 \end{pmatrix}$$

Problem 2. Normalization Using Python

Check using Google Colab. You can use the scikit-learn library to do so:

```
1 import numpy as np
2
3 v1 = np.array([[2], [1]])
4 print(v1)
5
6 from sklearn.preprocessing import normalize
7
8 v2 = v1 / np.linalg.norm(v1)
9 print(v2)
```

Solution. After running the provided Python code, the output is:

$$|\vec{v}_{1\text{normalized}}\rangle \approx \begin{pmatrix} 0.8944 \\ 0.4472 \end{pmatrix}$$

Problem 3. Normalization by Hand with Complex Coefficient

Normalize vector $|\vec{v}_1\rangle = 2|\uparrow\rangle + (1 + i)|\downarrow\rangle$ by hand.

Solution. The norm of $|\vec{v}_1\rangle = 2|\uparrow\rangle + (1 + i)|\downarrow\rangle$ is calculated as follows:

$$||\vec{v}_1\rangle| = \sqrt{|2|^2 + |1 + i|^2} = \sqrt{4 + (1^2 + 1^2)} = \sqrt{4 + 2} = \sqrt{6}$$

Thus, the normalized vector is:

$$|\vec{v}_{1\text{normalized}}\rangle = \frac{2}{\sqrt{6}}|\uparrow\rangle + \frac{1 + i}{\sqrt{6}}|\downarrow\rangle \approx \begin{pmatrix} 0.8165 \\ 0.4082 + 0.4082i \end{pmatrix}$$

Problem 4. Normalization Using Python with Complex Coefficient

Verify using Google Colab. Remember you need to use $1 + 1j$ to represent $1 + i$.

Solution. Using the following Python code:

```
1 v1 = np.array([[2], [1+1j]])
2 v2 = np.round(v1 / np.linalg.norm(v1), 4)
3 print(f"Normalized vector:\n{v2}")
```

After running the provided Python code, the output is:

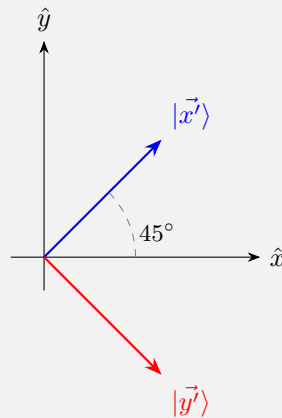
$$|v_{1\text{normalized}}\rangle \approx \begin{pmatrix} 0.8165 \\ 0.4082 + 0.4082i \end{pmatrix}$$

Problem 5. Vectors on a 2D Plane

On a 2D plane, draw the vectors $|\vec{x'}\rangle = \frac{1}{\sqrt{2}}(|\hat{x}\rangle + |\hat{y}\rangle)$ and $|\vec{y'}\rangle = \frac{1}{\sqrt{2}}(|\hat{x}\rangle - |\hat{y}\rangle)$. Write the column forms of $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$. Show that $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ are orthonormal, i.e. $\langle \vec{x'} | \vec{x'} \rangle = \langle \vec{y'} | \vec{y'} \rangle = 1$ and $\langle \vec{x'} | \vec{y'} \rangle = \langle \vec{y'} | \vec{x'} \rangle = 0$ and . You have derived a new orthonormal basis for a 2D plane. But this is not surprising, right? As $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ can be obtained by just rotating

Solution. The column forms of $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$, using $|\hat{x}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\hat{y}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, are

$$|\vec{x'}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\vec{y'}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$



Norms:

$$\langle \vec{x'} | \vec{x'} \rangle = \frac{1}{2} (1 \cdot 1 + 1 \cdot 1) = \frac{2}{2} = 1, \quad \langle \vec{y'} | \vec{y'} \rangle = \frac{1}{2} (1 \cdot 1 + (-1) \cdot (-1)) = \frac{2}{2} = 1$$

Cross terms:

$$\langle \vec{x'} | \vec{y'} \rangle = \frac{1}{2} (1 \cdot 1 + 1 \cdot (-1)) = \frac{1-1}{2} = 0, \quad \langle \vec{y'} | \vec{x'} \rangle = \frac{1}{2} (1 \cdot 1 + (-1) \cdot 1) = \frac{1-1}{2} = 0$$

Since both vectors have unit norm and their inner product with each other is zero, $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ form an orthonormal basis.

This is not surprising: $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ are simply $|\hat{x}\rangle$ and $|\hat{y}\rangle$ rotated counterclockwise by 45° , and a rotation always preserves lengths and angles, so it necessarily maps an orthonormal basis to another orthonormal basis.

Problem 6. Basis Change by Hand

Vector $\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is in the $|+\rangle / |-\rangle$ basis. Represent it in the $|0\rangle / |1\rangle$ basis. Try to convert $|+\rangle$ and $|-\rangle$ in terms of $|0\rangle$ and $|1\rangle$ and perform the substitution. To check your answer, convert it back to $|+\rangle / |-\rangle$ using the equations introduced in this chapter and you should get back $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

Solution. We have $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Substituting $\vec{v}_1 = 1|+\rangle + 2|-\rangle$:

$$\begin{aligned} \vec{v}_1 &= 1|+\rangle + 2|-\rangle \\ &= 1 \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + 2 \cdot \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}[(1+2)|0\rangle + (1-2)|1\rangle] \\ \vec{v}_1 &= \frac{3}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \end{aligned}$$

So in the $|0\rangle / |1\rangle$ basis:

$$\vec{v}_1 = \begin{pmatrix} 3/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \approx \begin{pmatrix} 2.1212 \\ -0.7071 \end{pmatrix}$$

Check. Converting back to the $|+\rangle / |-\rangle$ basis with $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, and writing $\vec{v}_1 = a|0\rangle + b|1\rangle$ with $a = 3/\sqrt{2}$, $b = -1/\sqrt{2}$:

$$\begin{aligned} c_+ &= \frac{a+b}{\sqrt{2}} = \frac{3/\sqrt{2} - 1/\sqrt{2}}{\sqrt{2}} = \frac{2/\sqrt{2}}{\sqrt{2}} = 1 \\ c_- &= \frac{a-b}{\sqrt{2}} = \frac{3/\sqrt{2} + 1/\sqrt{2}}{\sqrt{2}} = \frac{4/\sqrt{2}}{\sqrt{2}} = 2 \end{aligned}$$

which recovers $\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in the $|+\rangle / |-\rangle$ basis, exactly as expected.

Problem 7. Vector Normalization by Hand and Using Python

Normalize vector $|\vec{v}_1\rangle = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (which is in the $|0\rangle / |1\rangle$ basis) in the $|+\rangle / |-\rangle$ basis. Use Google Colab to verify your answer.

Solution. From Problem 6, $\vec{v}_1$ expressed in the $|+\rangle / |-\rangle$ basis (before normalization) is

$$\vec{v}_1 = \begin{pmatrix} 3/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

Its norm is

$$|\vec{v}_1| = \sqrt{\left(\frac{3}{\sqrt{2}}\right)^2 + \left(\frac{-1}{\sqrt{2}}\right)^2} = \sqrt{\frac{9}{2} + \frac{1}{2}} = \sqrt{5}$$

so the normalized vector, in the $|+\rangle / |-\rangle$ basis, is

$$|\vec{v}_{1\text{normalized}}\rangle = \frac{3/\sqrt{2}}{\sqrt{5}}|+\rangle - \frac{1/\sqrt{2}}{\sqrt{5}}|-\rangle = \frac{3}{\sqrt{10}}|+\rangle - \frac{1}{\sqrt{10}}|-\rangle \approx \begin{pmatrix} 0.9487 \\ -0.3162 \end{pmatrix}$$

Using Python to verify:

```

1 import numpy as np
2
3 # v1 = (1, 2) in the |0>/|1> basis
4 v1_01 = np.array([[1], [2]])
5
6 # basis change matrix from |0>/|1> to |+>/|-> coefficients
7 M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])
8
9 v1_pm = np.round(M @ v1_01, 4)
10 print("v1 in +/- basis (unnormalized):", v1_pm.ravel())
11
12 v1_pm_normalized = np.round(v1_pm / np.linalg.norm(v1_pm), 4)
13 print("v1 in +/- basis (normalized):", v1_pm_normalized.ravel())

```

which prints $\begin{pmatrix} 0.9487 \\ -0.3162 \end{pmatrix}$, matching the hand-calculated result.

Problem 8. Measurement Probability

For the $\vec{v}_1$ in the last question, what is the probability to find $|+\rangle$ in the $|+\rangle / |-\rangle$ basis measurement?

Solution. The probability of measuring $|+\rangle$ is the squared magnitude of its coefficient in the normalized vector:

$$P(|+\rangle) = \left| \frac{3}{\sqrt{10}} \right|^2 = \frac{9}{10} = 0.9$$

Problem 9. Vector Normalization by Hand and Using Python 2

Normalize $|\vec{v}_1\rangle$ in the $|0\rangle / |1\rangle$ basis. What is the probability to find $|0\rangle$ in the $|0\rangle / |1\rangle$ basis measurement? Use Google Colab to verify your answer.

Solution. The norm of $\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in the $|0\rangle / |1\rangle$ basis is

$$|\vec{v}_1| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

so the normalized vector is

$$|\vec{v}_{1\text{normalized}}\rangle = \frac{1}{\sqrt{5}} |0\rangle + \frac{2}{\sqrt{5}} |1\rangle \approx \begin{pmatrix} 0.4472 \\ 0.8944 \end{pmatrix}$$

The probability to find $|0\rangle$ is

$$P(|0\rangle) = \left| \frac{1}{\sqrt{5}} \right|^2 = \frac{1}{5} = 0.2$$

Using Python to verify:

```

1 import numpy as np
2
3 v1 = np.array([[1], [2]])
4 v1_normalized = v1 / np.linalg.norm(v1)
5 print("v1 normalized in 0/1 basis:", v1_normalized.ravel())
6
7 prob_0 = np.abs(v1_normalized[0, 0]) ** 2
8 print("P(|0>):", prob_0)

```

which prints $\begin{pmatrix} 0.4472 \\ 0.8944 \end{pmatrix}$ and $P(|0\rangle) = 0.2$, matching the hand-calculated result.

Problem 10. Basis Change and Measurement

If I get $|1\rangle$ in the measurement in Problem 9, what is the probability of getting $|+\rangle$ if I measure again in the $|+\rangle / |-\rangle$ basis measurement? (hint: after getting $|1\rangle$, how to represent it in the $|+\rangle / |-\rangle$ basis?)

Solution. After the measurement collapses the state to $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in the $|0\rangle / |1\rangle$ basis, we convert it to the $|+\rangle / |-\rangle$ basis using $a = 0$, $b = 1$:

$$c_+ = \frac{a+b}{\sqrt{2}} = \frac{0+1}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad c_- = \frac{a-b}{\sqrt{2}} = \frac{0-1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

so $|1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle$, which is already normalized. The probability of getting $|+\rangle$ is

$$P(|+\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} = 0.5$$

APPENDIX 1

Assignment 02 | Code Solution

assignment-02-code

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1 Assignment II

1.1 Import External Libraries

```
[1]: import numpy as np

from sklearn.preprocessing import normalize
```

1.2 2. Normalization Using Python

```
[2]: v1 = np.array([[2], [1]])
print(v1)

v2 = v1 / np.linalg.norm(v1)
print(v2)
```

```
[[2]
 [1]]
[[0.89442719]
 [0.4472136 ]]
```

1.3 4. Normalization Using Python with Complex Coefficient

```
[3]: v1 = np.array([[2], [1+1j]])
v2 = v1 / np.linalg.norm(v1)
print(v2)
```

```
[[0.81649658+0.j          ]
 [0.40824829+0.40824829j]]
```

1.4 7. Vector Normalization by Hand and Using Python

```
[4]: # v1 = (1, 2) in the |0>/|1> basis
v1_01 = np.array([[1], [2]])

# basis change matrix from |0>/|1> to |+>/|-> coefficients
M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])

v1_pm = M @ v1_01
print("v1 in +/- basis (unnormalized):", v1_pm.ravel())

v1_pm_normalized = v1_pm / np.linalg.norm(v1_pm)
print("v1 in +/- basis (normalized):", v1_pm_normalized.ravel())
```

```
v1 in +/- basis (unnormalized): [ 2.12132034 -0.70710678]
v1 in +/- basis (normalized): [ 0.9486833  -0.31622777]
```

1.5 9. Vector Normalization by Hand and Using Python 2

```
[5]: v1 = np.array([[1], [2]])
v1_normalized = v1 / np.linalg.norm(v1)
print("v1 normalized in 0/1 basis:", v1_normalized.ravel())

prob_0 = np.abs(v1_normalized[0, 0]) ** 2
print("P(|0>):", prob_0)
```

```
v1 normalized in 0/1 basis: [0.4472136  0.89442719]
P(|0>): 0.19999999999999998
```
